Introduction to Finance
Capital Budgeting (Part II)

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Contents

1 Why are cash flows central?

In project evaluation, the fundamental principle is that investment decisions must be based on cash flows rather than accounting profits.

The economic objective of a firm is to create value, and that value is measured by comparing the incremental cash flows generated by a project with the cost of the investment, using criteria such as Net Present Value (NPV).

The rule is simple in appearance:

> *A project adds value if the present value of the cash flows it generates exceeds the initial investment cost.*

However, correctly identifying those cash flows is one of the most complex—and most important—aspects of project evaluation. The difficulty lies not in the NPV technique itself, but in the accurate construction of the flows to be discounted.

A central principle is the separation between investment decisions and financing decisions:

> *Project cash flows are calculated as if the project were funded entirely with equity; the effect of financing (debt, equity, or a combination thereof) is reflected in the discount rate, not in the cash flows.*

For this reason, debt interest, principal repayments, and dividends are not part of the project's cash flows. These represent how the flows generated by the investment are distributed between creditors and shareholders, and their effect is incorporated indirectly through the discount rate.

> *Consequently, project evaluation does not consist of projecting accounting results, but of determining whether the cash flows generated by an investment are sufficient to compensate for the risk assumed and the opportunity cost of capital. This is the logic underlying the NPV criterion.*

Question: Why, in the evaluation of energy projects, is it necessary to incorporate assumptions regarding technological evolution and market structure when projecting cash flows? Justify your answer using evidence from the solar energy sector.

2 Incremental cash flows

2.1 The Question is Not "How Much Does the Project Earn?"

The relevant cash flows for project evaluation are incremental cash flows, defined as the difference between the cash flows the firm generates with the project and those it would generate without the project. A flow is relevant only if it is incremental, occurs within the evaluation horizon, and effectively impacts the project's cash position.

Incremental Cash Flow = Cash Flow with Project - Cash Flow without Project

Therefore, the correct question is never *How much does the project earn?* but rather:

> *How do the firm's total cash flows change if the project is accepted? In other words, the goal is to compare two alternative worlds: the state of the firm if the project is accepted versus the state of the firm if the project is rejected. Only the differences between these two worlds are economically relevant.*

This principle is key to determining which flows should be included and which should be excluded from the evaluation.

2.2 Common Errors of Inclusion and Exclusion in Cash Flows

2.2.1 Sunk Costs

A sunk cost is an outlay made in the past that cannot be recovered and does not vary regardless of whether the decision is made to accept or reject the project.

Typical examples include market studies already paid for or research and development expenses incurred previously.

> *Sunk costs are not included in project cash flows. Including them can lead to rejecting projects that, from an economic standpoint, actually create value.*

The exclusion of sunk costs is not an arbitrary convention, but rather a direct consequence of incremental cash flow analysis.

Question: What is the sunk cost fallacy? Check out this article. Fallacy of Concorde

Example 1:

If the company spent $50 million on a market study last year that is being used as an input for the project evaluation, should it be included in the project's cash flows?

This amount should not be included in the project's cash flows, as it was incurred regardless of the decision to accept or reject the project.

Example 2:

A tech company has invested US$ 800,000 over the last two years in developing a new App. When the project is at 80% development, a competitor launches a similar App that drastically reduces the company's market potential. They are evaluating whether or not to abandon the project. The CFO says: "We cannot abandon now; if we do, we will have thrown US$ 800,000 in the trash, and that will look terrible on the balance sheet." Do you agree with the CFO's opinion?

No, I do not agree with the CFO's opinion. The CFO is falling victim to the Sunk Cost Fallacy.

In financial decision-making, the US$ 800,000 already spent is a sunk cost: it is gone regardless of whether the company finishes the app or cancels it today. Decisions should be based on incremental costs and future benefits.

2.2.2 Opportunity Costs

An opportunity cost corresponds to the value of the best alternative use of a resource employed in the project.

> *Opportunity costs must be included in cash flows, even if they do not involve an explicit monetary outlay. Ignoring opportunity costs is one of the most frequent errors in project evaluation practice.*

Example 3:

It is often believed that using a resource the company already owns and does not use (such as a server or an empty office) has no cost. However, its opportunity cost is its value in its best alternative use (sale or lease), not zero.

Example 4:

If a company-owned plot of land that could be sold today for $200 million is integrated into a project, should it be included in the project costs?

This value represents an opportunity cost of the project, even though no actual cash payment is made.

2.2.3 Cannibalization (Negative Externality)

A project can reduce the cash flows of other products, services, or business lines within the company. This effect is known as cannibalization.

A typical example is the launch of a new product that reduces the sales of an existing one.

> *When evaluating a project's net impact, any lost cash flows from existing products must be deducted. However, if those losses are inevitable—such as when a competitor is poised to launch a rival product—the cannibalization should not be subtracted, as it does not represent an incremental loss caused by the project itself.*

Example 5:

A new product generates its own sales of \$500, but reduces the sales of an existing product by \$120. What is the relevant revenue for the project?

The relevant incremental revenue of the project is $500 - 120 = 380$.

2.2.4 Positive Externalities

A project can generate additional benefits for other areas or businesses of the company, such as improvements in brand image, an increase in customer traffic, etc.

> *Positive externalities should be incorporated into cash flows only if they can be measured in a reasonable way and are attributable to the project. Otherwise, it is preferable to treat them qualitatively to avoid overestimating the project's value.*

> ***Project evaluation is about correctly identifying the real changes in cash flows that an investment decision triggers within the company.***

3 Types of Cash Flows

Project Cash Flow (Free Cash Flow). This corresponds to the cash flow generated by the investment itself, considering only the income and expenses associated with the project's operation. This flow allows for the evaluation of the project's economic profitability, regardless of how it is financed; therefore, it does not include interest, debt repayments, or capital contributions. It is the flow used to calculate the project's NPV and IRR.

Equity Cash Flow (Investor Cash Flow). This corresponds to the cash flow received by equity contributors once the effects of financing are incorporated. It includes capital outlays, debt service, and the net benefits perceived by investors. This flow allows for the evaluation of the return on equity (ROE) and is used to calculate the investor's NPV and IRR.

Question: Is the same discount rate used to evaluate the project cash flow and the investor cash flow? If the answer is negative, what rate is used in each case?

> *The decision to invest is based on the project cash flow; the method of financing is evaluated using the investor cash flow.*
>
> *The project NPV determines if the investment creates value; the investor NPV determines how that value is distributed according to the financing structure.*

The separation between investment and financing decisions implies that a project must be evaluated by its ability to generate flows, and not by the way these are distributed between creditors and shareholders. The financing structure does not turn a bad project into a good one.

4 Evaluation Horizon

The evaluation horizon corresponds to the period during which the project's cash flows are explicitly estimated. It does not necessarily coincide with the project's total economic life, but rather with the timeframe for which it is possible to make projections with a reasonable degree of certainty and economic consistency. The evaluation horizon should not be confused with the physical life of assets nor with the depreciation period, but with the period for which flows can be estimated with a reasonable degree of reliability.

Determining the evaluation horizon is an analytical decision that depends on the nature of the project and the environment in which it develops. Among the main factors that influence this are:

Technological Obsolescence: When the project faces rapid technological change, the evaluation horizon is usually shorter, as future flows become highly uncertain.

Stability of the Environment: The credibility of projections decreases in unstable political, economic, or social contexts. This justifies a more limited evaluation horizon.

Contractual or Legal Restrictions: In projects subject to concessions, contracts, or temporary permits, the term of these legal/contractual instruments usually defines the maximum evaluation horizon.

Question: Search for examples of different types of projects and evaluation horizons.

When the evaluation horizon is shorter than the project's economic life, it becomes necessary to incorporate a continuation value that represents the remaining economic value of the project beyond the analyzed period.

5 Components of Cash Flows

5.1 Investments

Investments can be made *prior to start-up or during operation.*

Pre-operating investments. These include:

- Investments in fixed assets

- Investments in intangible assets
- Investments in working capital

Investments during operation. These include:

- Replacement investments
- Expansion investments

5.1.1 Pre-operating investments

The following flows are considered at $t = 0$.

5.1.1.1 Investments in fixed assets

These are investments in tangible assets used in the transformation process of inputs or that serve as support for the normal operation of the project.

This includes:

- Land
- Buildings
- Plant, office, and sales room equipment (machinery, furniture, tools, vehicles, etc.)
- Support service infrastructure (potable water, drainage, electrical grid, etc.)

Note: With the exception of land, all other fixed assets are subject to depreciation.

5.1.1.2 Investments in intangible assets

These are all investments made in services or acquired rights necessary for the start-up of the project. This includes:

- Organizational expenses (design of information systems and administrative procedures, legal expenses for the legal incorporation of the company).
- Start-up expenses (payments for salaries, rent, advertising, insurance, etc., made before the start of operations).
- Patents and licenses involving, among others, payments for the right to use a brand, formula, or production process, municipal permits, and notary authorizations.
- Pre-operating personnel training.
- Website development.

- Software.
- Contingencies (generally calculated as a percentage of total investments).

Notes:

- Project study costs should not be considered part of the investment because they are sunk costs that must be paid regardless of the evaluation outcome, making them irrelevant.*
- The accounting loss of value of intangible assets is called amortization.

> ***Depreciation and amortization are not cash flows, but they affect cash flows via taxes.***

5.1.1.3 Investments in working capital

Net working capital corresponds to the difference between current assets and operating current liabilities.

A project typically considers: inventory, accounts receivable, and accounts payable.

Working capital is defined as:

$$Working\ capital\ =\ Current\ assets\ -\ Current\ liabilities$$

> *Working capital does not measure profitability, but liquidity: it determines how much cash must be tied up so that the project can operate without interruptions.*

Several methods for calculating the investment in working capital are described below.

A. Accounting method (or projected balance sheet method)

The investment in working capital is calculated as the difference between the project's operating current assets and operating current liabilities, using pro forma balance sheets.

Question: What are pro forma financial statements?

It requires determining the optimal amounts of:

Current assets: (i) cash, (ii) inventories, and (iii) accounts receivable.

Current liabilities: (i) accounts payable to suppliers and (ii) short-term bank loans.

It is recommended for use in feasibility studies when complete financial projections are available.

B. Lag period method (or cash cycle method)

It calculates how many days of operating costs must be financed between the time the first payment is made for the acquisition of inputs and the collection of payment for the sale of products.

The investment in working capital (IWC) is calculated as:

$$IWC = \frac{C_a}{365} \times n_d$$

where C_a is the annual cash cost and n_d is the number of lag days (cash cycle). Annual cash costs represent the operating outlays that the project must pay in cash and, therefore, determine the magnitude of the investment in working capital.

This method is applied in pre-feasibility studies or when projected balance sheets are unavailable. It requires a good estimation of the time lag.

Example 6:

- Suppliers grant 30-day credit for the payment of inputs.
- Inventories remain for 20 days.
- Sales are collected 40 days later.

If the annual cash costs are \$1095, calculate the investment in working capital.

Total lag $= 20 + 40 - 30 = 30$ days

$$IWC = \frac{1095}{365} \times 30 = 90$$

How does your answer change if the inputs must be paid for at the time of purchase?

Total lag $= 20 + 40 - 0 = 60$ days

$$IWC = \frac{1095}{365} \times 60 = 180$$

Supplier credit reduces the need for working capital by half.

C. Maximum cumulative deficit method

Actual cash inflows and outflows are projected at the time they are collected or paid (weekly or monthly), as the goal is to measure real liquidity needs. The investment in working capital is defined as the largest cumulative cash deficit that appears during the period.

The investment in working capital depends on the timing and the manner in which the operation generates and collects its flows; therefore, different methods reflect different levels of information and precision.

- Early stage / limited data: Lag Period Method
- Complex or seasonal project: Maximum Cumulative Deficit Method
- Mature project with pro forma statements: Accounting Method

5.1.2 Investments during operation

These flows can occur at $t = 1, 2, ... T$.

5.1.2.1 Replacement investment

These are carried out due to: (a) insufficient capacity of current equipment, (b) increased maintenance costs due to the age of the machine, (c) decreased productivity due to increased downtime for maintenance reasons, or (d) technological obsolescence.

Additionally, if the project replaces an existing asset, the following must be considered:

- The sale value of the old asset.
- Taxes on capital gains or losses.

5.1.2.2 Expansion investments

They may also require an increase in the investment amount for working capital.

Example 7:

- Inventories increase by 80
- Accounts receivable increase by 40
- Accounts payable increase by 30

The change in working capital is:

$$\Delta WC = 80 + 40 - 30 = 90$$

> **An increase in working capital is a use of cash (negative flow).**
>
> **A reduction in working capital is a source of cash (positive flow).**

5.2 Revenues

5.2.1 Operating revenues

These are the revenues generated by the project's primary activity, derived from the sale of goods or the provision of services.

- They are included net of VAT (Value Added Tax) when it is recoverable.
- They are recorded in the period in which they are economically generated.
- They must reflect only the incremental effect of the project.

Examples include: sales of manufactured products, income from services rendered, and fees charged for the use of infrastructure.

Question: If a sale occurs in Month 2, but the cash is received in Month 5, what is the effect on the monthly cash flows? What is the effect on the investment in working capital?

5.2.2 Revenue from the sale of assets

These correspond to the income obtained from the disposal of assets associated with the project, whether during its operation or at the end of the evaluation horizon.

- They are considered net of taxes.
- The relevant factor is the capital gain or loss, calculated relative to the asset's book value.
- They are especially important in the terminal cash flow.

> *When an asset is sold, taxes are paid on the capital gain—the difference between the sale price and its book value. Conversely, if the book value exceeds the market value, the resulting loss creates a tax shield, effectively saving the company money on taxes.*

Example 8:

Suppose a company decides to sell a machine in year 8 of operation with the following data:

- Sale Price (Market Value): $2,500,000
- Original Acquisition Cost: $10,000,000
- Accounting useful life: 10 years (straight-line depreciation of $1.000.000 per year)
- Book Value at the time of sale: $2.000.000 (corresponding to the 2 remaining years of depreciation)

- Income Tax Rate: 20%

Calculate the impact on the cash flow (CF).

$$\Delta CF = \text{Book value} + \text{Profit on sale} \times (1 - \text{Tax rate}) = 2000000 + 500000 \times (1 - 0.2) = 2400000$$

$$\Delta CF = \text{Sales value} - (\text{Sale value} - \text{Book value}) \times \text{Tax rate} = 2500000 - 500000 \times 0.2 = 2400000$$

$$\Delta CF = 2500000 - 2000000 \times 0.2 = 2400000$$

Question: Calculate the cash flow impact assuming the machine is sold at its book value.

Note: The book value is calculated as the original purchase price minus all the depreciation deductions taken up to that point.

5.3 Manufacturing costs and operating expenses

These correspond to the outlays necessary for the project to operate and generate revenue.

In project evaluation, only incremental and relevant costs should be included, regardless of their accounting classification.

5.3.1 Direct and indirect manufacturing costs

This includes costs associated with the production process prior to the marketing and administration stages.

Direct Materials. Inputs that are physically incorporated into the final product and can be clearly identified per unit produced. For example: wood in furniture manufacturing, steel in metal structures, and flour in bread production.

Direct Labor. Corresponds to the compensation and disbursements associated with personnel directly involved in production. It includes wages and salaries, social security contributions, severance pay, bonuses, and other payments directly linked to productive work.

Indirect Materials. Inputs necessary for the production process that are not directly incorporated into the final product or cannot be assigned to a specific unit. This includes: spare parts, fuel and lubricants, cleaning supplies, maintenance materials, etc.

Indirect Labor. Compensation and disbursements associated with personnel who support the production process but do not participate directly in the manufacturing of the good. It includes disbursements related to the salary or wage of production managers, maintenance staff, internal drivers, cleaning staff, and security guards.

Manufacturing overhead (Indirect manufacturing expenses). Costs necessary for the operation of the production plant that cannot be assigned directly to a unit produced. These include, among others: energy (electricity, gas, steam), telephone expenses, insurance, leases of production facilities, and depreciation of productive assets.

5.3.2 Operating expenses

These correspond to outlays associated with administration and marketing, which are necessary for the project's operation but are not directly linked to the production process.

General and Administrative Expenses (G&A). This includes labor costs for administrative personnel, entertainment expenses, insurance, office rentals, office materials and supplies, depreciation of administrative buildings and office equipment, taxes, etc.

Selling Expenses. These are the expenses associated with the marketing and distribution of products or services. They include: compensation for sales staff, sales and collection commissions, advertising costs, packaging, transportation, and warehousing.

5.4 Other expenses

Financial expenses. These include interest on loans obtained to finance the project. In the project's economic evaluation, these expenses should not be included in the cash flows. They are only included when constructing the investor's cash flow.

Provision for uncollectible accounts and contingency allowances. These correspond to provisions or estimates of potential losses, usually calculated as a percentage of expenses or sales.

- If they represent expected actual cash outflows, they should be included.
- If they are merely accounting provisions, they should not be incorporated into the cash flows.

5.5 Continuation value

Continuation value (also called terminal value) is the estimated value of the project's free cash flows beyond the explicit forecast horizon. It assumes that at the end of the forecast period (T), the project or firm will not be liquidated but will continue to operate as a "going concern" in the long run.

It can be calculated using a perpetuity formula if the project's cash flows are expected to grow at a constant rate (g) forever:

$$V_T = \frac{FCF_{T+1}}{r - g}$$

- In a growing perpetuity, the cash flow must account for the fact that a portion of earnings is reinvested. It is commonly assumed that, in the long run, capital expenditures (CapEx) roughly equal depreciation to maintain productive capacity, or the flow is adjusted to reflect the net investment required to sustain the growth rate g.
- In this scenario, net working capital is not recovered as a single cash inflow at the end of the horizon (T), as it remains tied up in the ongoing operations of the business.

5.6 Liquidation/salvage value

This represents the net cash flow received when a project is shut down and its assets are disposed of. This approach is used for projects with a finite life where assets are no longer needed and are sold, either for their resale value or as scrap.

The after-tax salvage value is calculated by accounting for the tax (τ) impact of any "gain" or "loss" on the sale relative to the asset's book value.

$$\text{After-Tax Salvage Value} = \text{Sale Price} - \tau \times (\text{Sale Price} - \text{Book Value})$$

where:

$$\text{Gain on Sale} = \text{Sale Price} - \text{Book Value}$$

- The Book Value is the original purchase price minus all accumulated depreciation that has been deducted for tax purposes.
- Unlike continuation value, in a liquidation scenario, the recovery of net working capital is included as a final cash inflow (e.g., collecting outstanding accounts receivable and selling off remaining inventory).

Example 9:

If the net cash flow were \$12000, depreciation was \$2000, and the cost of capital rate was 10%, the continuation value would be:

$$\frac{12000 - 2000}{0.1} = 100000$$

Example 10:

A company is considering opening new retail stores. The free cash flow projections for the new stores are shown below (in millions of dollars):

$I_0 = -10.5$, $CF_1 = -5.5$, $CF_2 = 0.8$, $CF_3 = 1.2$, $CF_4 = 1.3$

After year 4, the company expects free cash flow from the stores to increase at a rate of 5% per year. If the appropriate cost of capital for this investment is 10%, calculate the continuation value in year 4 and the NPV of the new stores.

- The continuation value is

$$\text{Continuation value} = \frac{CF_{T+1}}{r-g} = \frac{1.3 \times 1.05}{0.10 - 0.05} = 27.3$$

- The NPV is

$$NPV = -10.5 - \frac{5.5}{1.1} + \frac{0.8}{1.1^2} + \frac{1.2}{1.1^3} + \frac{1.3}{1.1^4} + \frac{27.3}{1.1^4} = 5.6$$

6 Inflation in Cash Flows

Cash flows can be projected either including or excluding inflation.

- If projected including inflation (nominal flows): They must be discounted using a nominal discount rate.
- If projected in constant purchasing power (real flows): They must be discounted using a real discount rate.

Example 11:

Consider a project with the following nominal cash flows: $I_0 = \$1000$, $CF_1 = \$600$, and $CF_2 = \$650$.

The nominal discount rate is 14%, and the expected inflation rate is 5%.

(a) Calculate the NPV using nominal cash flows

When using nominal flows, we simply discount them by the nominal rate:

$$NPV = -1000 + \frac{600}{1.14} + \frac{650}{1.14^2} = 26.47$$

(b) Calculate the NPV using real cash flows

First, we determine the real discount rate (r) using the following equation:

$$(1 + i) = (1 + \pi^E)(1 + r) \implies r = \frac{1.14}{1.05} - 1 = 0.0857$$

Next, we convert the nominal flows into real terms by deflating them by the expected inflation:

$CF_1 = \frac{600}{1.05} = 571.43$

$CF_2 = \frac{650}{1.05^2} = 589.57$

Now, we calculate the NPV by discounting these real flows at the real discount rate:

$$NPV = -1000 + \frac{571.43}{1.0857} + \frac{589.57}{1.0857^2} \approx 26.47$$

The NPV remains identical regardless of whether you use nominal or real terms, provided the cash flows and discount rates are treated consistently.

Question: What is the effect on the NPV of using a nominal rate to discount real flows?

7 VAT in Cash Flows

Project cash flows should record sales exclusive of Value Added Tax (VAT). Because VAT is a tax collected on behalf of the government rather than earned revenue, including it would inaccurately inflate the project's actual inflows.

However, it must be incorporated into the calculation of working capital investment if it is paid on purchases before it can be recovered.

8 Project Cash Flow Construction

The project's cash flow for the period $1, 2, \ldots, T$ is calculated as:

Component
Revenues
(-) Cost of goods sold (COGS)
= Gross Profit
(-) Sales and administrative expenses
= EBITDA
(-) Depreciation
(-) Amortization
= EBIT
(-) Taxes (= EBIT x Tax rate)
= Unlevered net income
(+) Depreciation
(+) Amortization
(-) Capital expenditures /Other investments
(-) Changes in working capital
(+) Liquidation/Continuation value
= Free cash flow

EBIT (**E**arnings **B**efore **I**nterests and **T**axes)

EBITDA (**E**arnings **B**efore **I**nterest, **T**axes, **D**epreciation and **A**mortization)

Note that:

$$EBITDA = EBIT + Depreciation + Amortization$$

- *Unlevered net income* (also referred to as *Net Operating Profit After Tax* or $NOPAT$) is a measure of a project's or firm's profitability that excludes the impact of financing decisions. It represents the net income the project would generate if the firm had no debt (i.e., it is "unlevered"). By excluding interest expenses, it allows financial managers to evaluate a project's performance based solely on its own merits, separate from how the firm chooses to pay for it. It allows for a "fair" comparison between different projects or companies, regardless of whether one is financed with 100% equity and another is heavily burdened with debt.
- *Changes in working capital:* This represents the "liquidity" required to run the project. An increase in WC is a cash outflow.

$$\Delta WC = WC_t - WC_{t-1}$$

At $t = 0$: Usually a cash outflow (initial investment).

At $t = T$ (Liquidation): Usually a cash inflow (recovery of working capital).

- *Liquidation/Continuation value:* If you are liquidating, remember to include the after-tax salvage value of the assets and the recovery of working capital.

Example 12:

A company is in the following situation:

- Operating Revenue: 1000
- Operating Costs (cash): 600
- Depreciation: 100
- Tax Rate: 30%

Calculate the EBIT and the project's cash flow.

$$EBIT = 1,000 - 600 - 100 = 300$$

The project cash flow (CF) is:

$$CF = 300 \times (1 - 0.3) + 100 = 310$$

9 Additional questions

Question 1

The startup “Glamping Tech” aims to install a complex of 5 smart domes in Torres del Paine. The project evaluation horizon is 3 years.

1. Initial Investment ($t = 0$):

- Fixed Assets: \$2,000,000. These are depreciated linearly over 5 years with a salvage value of zero at the end of their useful life.
- Working Capital (WC): It is estimated that the company must maintain a WC equivalent to 15% of the following year's total cash operating costs.

2. Projected Operations:

- Revenue: \$1,400,000 in Year 1, with an expected growth of 10% for Year 2. In Year 3, sales remain constant relative to Year 2.
- Cash Operating Costs: These represent 50% of sales for each period.

3. Salvage Value and Closing ($t = 3$):

- Sale of Assets: The domes will be sold at a market value of \$1,100,000.
- Recovery of Working Capital: 100% of the invested working capital is recovered.

4. Financial Parameters:

- Tax Rate: 27%
- Discount Rate (WACC): 12%

(a) Construct the Net Cash Flow for years 0, 1, 2, and 3.

(b) Calculate the NPV. Do you accept or reject the project?

(c) Calculate the payback period using non-discounted cash flows

Solution:

(a)

WC requirement	$105.000	$115.500	$115.500	$0

	0	1	2	3
Revenues		$1.400.000	$1.540.000	$1.540.000
Cash operating costs		$-700.000	$-770.000	$-770.000
Dome revenue				$1.100.000
Book value (domos)				$-800.000
Depreciation		$-400.000	$-400.000	$-400.000
Taxable earnings		$300.000	$370.000	$670.000
Taxes		$-81.000	$-99.900	$-180.900
Earnings after tax		$219.000	$270.100	$489.100
Depreciation		$400.000	$400.000	$400.000
Book value (domes)				$800.000
Investment in fixed assets	-$2.000.000			
Investment in working capital	$-105.000	$-10.500	$0	
Recovery of working capital				$115.500
Free cash flow	**-$2.105.000**	**$608.500**	**$670.100**	**$1.804.600**

Discount rate	12%
NPV	$256.982

Note: The net cash flow from the sale of the domes is equal to:

ΔCF = Book value + Profit on sale × (1-Tax rate) $= 800000 + (1100000 - 800000) \times 0.73 = 1019000$

or

ΔCF = Sales value − Profit on sale × Tax rate $= 1100000 - (1100000 - 800000) \times 0.27 = 1019000$

(b) The NPV is equal to:

$$NPV = -2105000 + \frac{608500}{1.12} + \frac{670100}{1.12^2} + \frac{1804600}{1.12^3} = 256982$$

Since this is a positive value, the “Glamping Tech” project is financially viable at a 12% discount rate.

(c) The investment is recovered in approximately 2.46 years

Period	Non-discounted cash flow	Cumulative
0	-2105000	-
1	608500	608500
2	670100	1278600
3	1804600	3083200

$$\text{Payback period} = 2 + \frac{826400}{1804600} \approx 2.46 \text{ años}$$

Question 2

You are considering expanding your business to manufacture a new type of packaging. To do so, you have two alternatives for acquiring a critical asset for the production process.

Alternative 1:

- For the first two years, you must rent a machine at a cost of \$500,000 per year.
- At the end of year 2, you purchase a top-tier machine for \$1,500,000.
- This machine has annual operating costs of \$240,000 and a useful life of 4 years.

Alternative 2:

- You purchase a machine available on the market today ($t = 0$) for \$990,000.
- Annual operating costs amount to \$400,000 over a useful life of 7 years.

Financial parameters:

- Depreciation: Both machines are fully depreciated over their useful lives.
- Tax rate: 0%
- Discount rate: 12%

Which alternative should you choose?

Solution:

Because the two alternatives have different useful lives and different timing for the cash outflows, we cannot simply compare their NPV. We must find which option has the lowest EAC.

Discount rate	**12%**
Tax rate	**0%**

Machine 1

	0	1	2	3	4	5	6
Tax savings from depreciation (tax shield)		0	0	0	0	0	0
Annual costs		\$500.000	\$500.000	\$240.000	\$240.000	\$240.000	\$240.000
Investment			\$1.500.000				

PVC	**\$2.621.942**
CAE	**\$637.724**

Machine 2

	0	1	2	3	4	5	6	7
Tax savings from depreciation (tax shield)		0	0	0	0	0	0	0
Annual costs		\$400.000	\$400.000	\$400.000	\$400.000	\$400.000	\$400.000	\$400.000
Investment	\$990.000							

PVC	**\$2.815.503**
CAE	**\$616.927**

Machine 1:

$PV_{rent} = \frac{500000}{1.12} + \frac{500000}{1.12^2} = \frac{500000}{0.12} \times (1 - \frac{1}{1.12^2}) = 845025.5$

$PV_{operation} = \frac{1}{1.12^2} \times \frac{240000}{0.12} \times (1 - \frac{1}{1.12^4}) = 581125$

$PV_{adquisition} = \frac{1500000}{1.12^2} = 1195791$

$PVC_{total} = 1195791 + 845026 + 581125 = 2621942$

$EAC_1 = 0.12 \times 2621942 \times \left(\frac{1.12^6}{1{,}12^6 - 1}\right) = 637724$

Machine 2:

$PV_{operation} = \frac{400000}{0.12} \times [1 - \frac{1}{1.12^7}] = 1825503$

$PVC_{total} = 990000 + 1825503 = 2815503$

$EAC_2 = 0.12 \times 2815503 \times \left(\frac{1.12^7}{1{,}12^7 - 1}\right) = 616927$

Alternative 2 is recommended, as it presents a lower EAC based on a 12% discount rate.

Question 3

After completing your undergraduate studies, you face a key career decision. You can either begin working immediately in Chile or invest in your future by pursuing a Master's degree in Finance abroad.

Project A: Start working in Chile

Project B: Study abroad and complete a Master's in Finance

(a) Suppose the internal rate of return (IRR) of starting work in Chile is higher than that of pursuing the Master's degree. Based on this information alone, would you choose to stay and work in Chile? Explain your reasoning.

(b) Construct a set of realistic cash flows for each alternative that would allow you to evaluate this decision. Your answer should explicitly consider:

- Tuition and living expenses while studying
- Foregone salary during the study period
- Expected salary after completing the Master's degree
- A reasonable time horizon for comparison

Use numerical values and clearly justify your assumptions.

(c) Based on your cash flows, how would you decide between the two projects? Clearly state and apply the appropriate decision rule.

(d) Now suppose that:

- You can only finance the Master's degree through a loan at a relatively high interest rate, and
- There is uncertainty about whether you will secure a high-paying job after graduation (e.g., only a 70% probability of obtaining the expected salary, and a 30% probability of earning a lower salary).

(i) How would these factors affect your decision? Be precise about the channels through which they operate.

(ii) Should the interest rate on the loan be incorporated into the project's cash flows or the discount rate when evaluating Project B? Clearly justify your answer

Question 4

A project requires an initial working capital investment of 120 at $t = 0$. During its useful life, the operating level remains constant. At the end of the evaluation horizon, the company will continue to operate the project, maintaining the same level of activity (there is no closure or liquidation).What is the correct treatment of working capital in the project's cash flow?

A. Include a recovery of 120 as an inflow in the final period, because working capital is always recovered at the end of the evaluation horizon.

B. Do not include any recovery, because working capital is a permanent investment as long as the project continues to operate.

C. Include a partial recovery equivalent to the accumulated depreciation of the working capital.

D. Include the recovery only if the project's NPV is positive.

E. Do not include working capital in the cash flow, because it does not correspond to an actual cash outlay.

Question 5

Assume that the project is evaluated over a five-year time horizon. Two machines are acquired with a useful life of 10 years. The acquisition cost of each machine is $1,000, and both are fully depreciated over their useful life. It is estimated that after five years, the market value of Machine 1 will be $650 and that of Machine 2 will be $400.

If the tax rate is 20%, calculate the effect on the cash flow for year 5.

Solution:

The effect on the cash flow for year 5 is:

	Machine 1	Machine 2	Total
Sales Revenue	650	400	1050
Book Value	-500	-500	-1000
Gain/Loss on Sale	150	-100	50
Tax (20%)	-30	20	-10
Net Profit after Sale	120	-80	40
Book Value	500	500	1000
Cash Flow	**620**	**420**	**1040**

Machine 1

$$\Delta CF = 500 + 150 \times (1 - 0.2) = 650 - 30 = 620$$

Machine 2

$$\Delta CF = 500 - 100 \times (1 - 0.2) = 400 + 20 = 420$$

Therefore, the salvage values (terminal values) for Machines 1 and 2 are $620 and $420, respectively.

Question 6

You are bidding on a contract to provide 5 modified trucks per year for 4 years (20 trucks total). To win, you must submit the lowest bid per truck while ensuring you still earn your required 20% return.

Cost and Investment Data:

- *Variable Costs:* Platform cost is \$15,000/unit; Modification cost is \$7,000/unit.
- *Fixed Costs:* Annual lease for facilities is \$36,000.
- *Initial Investment:*
  - Equipment costs \$150,000 (depreciated straight-line to zero over 4 years).
  - Net Working Capital (NWC): An initial investment of \$50,000 is required (fully recovered in Year 4).
- *Salvage Value:* The equipment can be sold for \$25,000 at the end of Year 4.
- *Tax Rate:* 21%.

Calculate the minimum bid price.

Solution:

Item	Year 0	Year 1	Year 2	Year 3	Year 4
Trucks		5	5	5	5
Modification cost (\$ per truck)		7000	7000	7000	7000
Truck platform (\$ per truck)		15000	15000	15000	15000

Item	Year 0	Year 1	Year 2	Year 3	Year 4
Sales		?	?	?	?
Costs					
(-) Modification costs		\$-35.000	\$-35.000	\$-35.000	\$-35.000
(-) Truck platforms		\$-75.000	\$-75.000	\$-75.000	\$-75.000
(-) Facilities (leasing)		\$-36.000	\$-36.000	\$-36.000	\$-36.000
(-) Depreciation		\$-37.500	\$-37.500	\$-37.500	\$-37.500
Total Costs		\$-183.500	\$-183.500	\$-183.500	\$-183.500
Earnings before taxes		?	?	?	?
(-) Taxes (21%)		?	?	?	?
Earnings after taxes		?	?	?	?
(+) Depreciation		\$37.500	\$37.500	\$37.500	\$37.500
Initial investment					
(-) Change in WC	\$-50.000				
(-) Capital Spending	\$-150.000				
Terminal value					
(+) Recovery WC					\$50.000
(+) Equipment					\$19.750
Cash Flow	\$-200.000	OCF	OCF	OCF	OCF + 69750

Note: OCF refers to Operating Cash Flow.

The minimum bid price is identified by solving for the price where the project breaks even, specifically where $NPV = 0$.

$$-200000 + \frac{OCF}{1.2} + \frac{OCF}{1.2^2} + \frac{OCF}{1.2^3} + \frac{OCF}{1.2^4} + \frac{69750}{1.2^4} = 0$$

$$\frac{OCF}{0.2} \times \left[1 - \frac{1}{1.2^4}\right] = 200000 - \frac{69750}{1.2^4}$$

$$OCF \times 2.588735 = 166362.8 \Rightarrow OCF = 64264$$

Therefore, we can set the following expression equal to the OCF:

$$(Sales - 183500) \times 0.79 + 37500 = 64264 \Rightarrow Sales = 217379$$

Given the requirement for five trucks annually, each unit will be priced at \$43476 (=217379/5). This pricing structure yields a return of just over 20%. Bidding any lower would result in a negative NPV.